Guardian

The Wheelchair Travel Guardian

A wheelchair user wants to go to Cardiff Central.

On a packed train the accessible space is occupied by someone and their luggage.

No staff respond. The doors are closing.

The train leaves without them. They go home.

This happens every day. It should never happen at all.

Meet Guardian

- A wheelchair-mounted AI travel guardian that manages your entire journey across the Welsh transport network by voice, automatically, in Welsh or English

The problems we set out to solve

No Unified Journey View

- Separate apps for different public transport
- Nothing shows the full trip in one place

Uncertainty Stops People Travelling

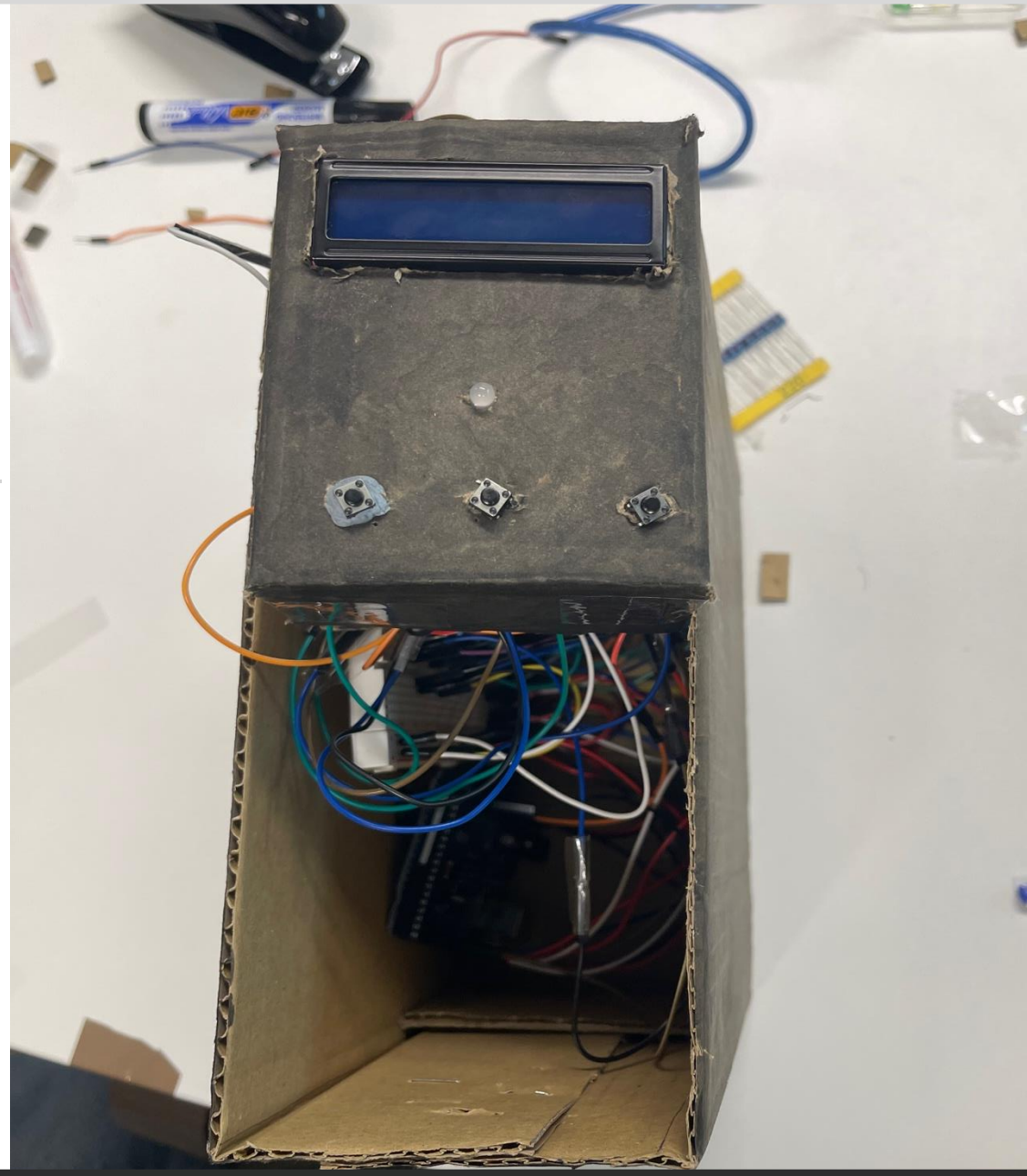
- Passengers do not know if space exists on the train
- Past bad experiences keep them wary of travel. Leading them to not take public transport.

Disruptions Leave Passengers Stranded

- Cancelled trains or buses = no guaranteed alternative
- Wheelchair users have no other options

How it works

- You tell the guardian where you want to go.
- It plans an accessible route across bus and/or train.
- It reserves your wheelchair space.
- If disruptions occur it finds an alternative instantly.



Four ways to travel with Guardian

Pre-planned Commute

Guardian runs protection automatically

No action required from the passenger

Advance Booking

Tell the guardian by voice where to go

It books everything for you

Spontaneously at station

Arrive with no booking

Guardian detects you and sets up the journey on the spot

Intent-Based Travel

You say what you want to do

Guardian does the rest.

Directly addressing the challenge

Smart connected station infrastructure

NFC readers extend TfW's existing contactless hardware

Seamless multimodal travel

Unifies one journey using your voice, for e.g bus, train and bus.

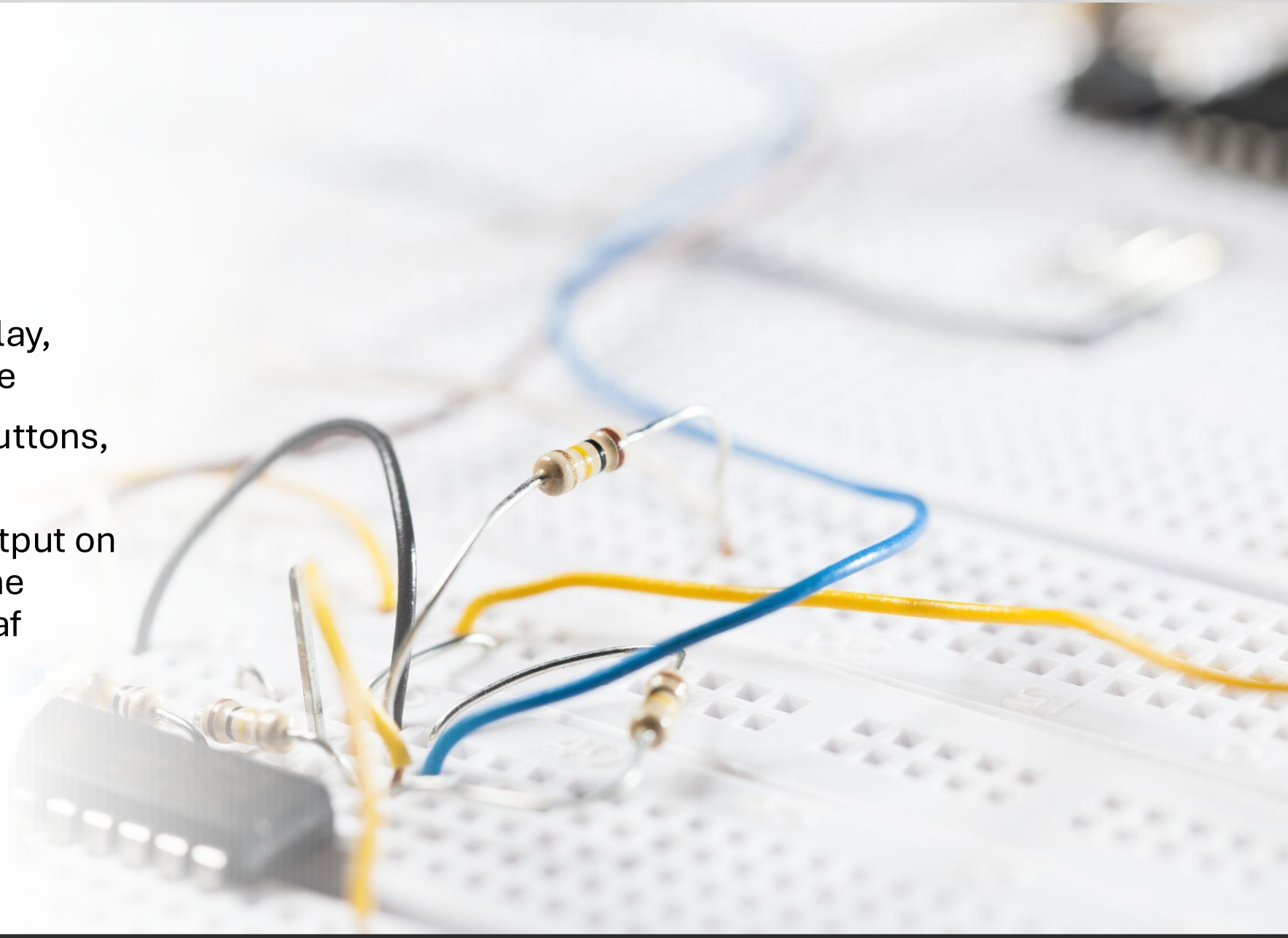
Retrofittable interactive components

Module clips to any wheelchair. NFC extends existing readers

What is the Module

Core Components

- Arduino brain, LCD display, speaker and microphone
- Status LEDs, physical buttons, alert buzzer
- LCD mirrors all voice output on to the screen which is the primary interface for deaf users

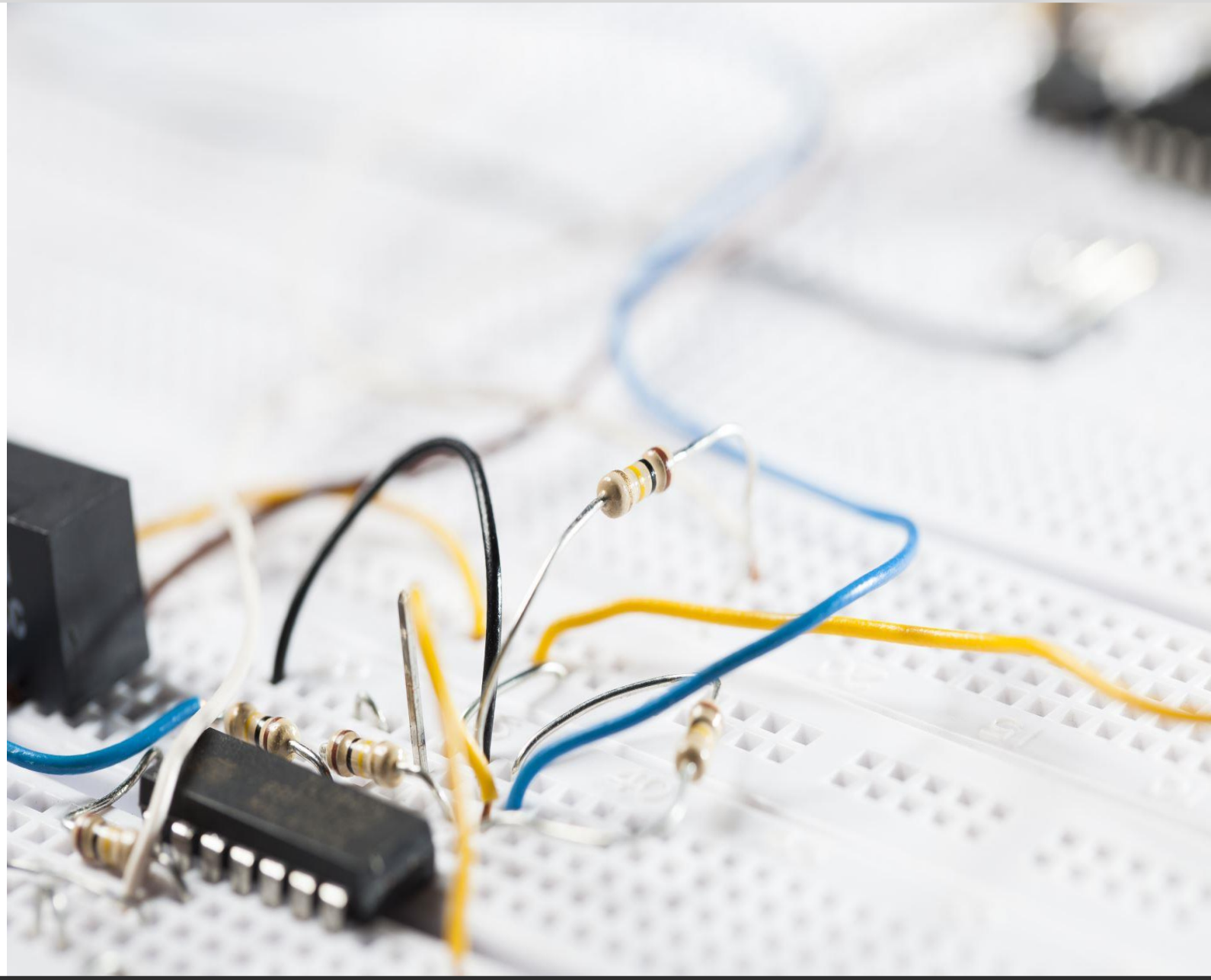


Production Features

- A shape-changing retractable screen that disappears when not in use.
- Will be IP67 waterproofed.
- Kinetic charging from the wheel hub. The ensures user never plugs anything in.

Why Adruino

- Arduino gives a dedicated, always-on, hands-free device. It's retrofittable



Why ElevenLabs

- ElevenLabs produces natural Welsh speech, not a robotic translation, which directly addresses the language requirement.
- It shows voice outputs which is the only interface that works for someone with limited hand mobility as moves around a lot.

Built On What TfW Already Has

FEIBR Fibre Network

Real-time module-to-train connectivity

No new data infrastructure needed

Live Timetable Data

Pulls TfW's existing feeds

Gets all departures, delays and cancellations

Passenger Assist

Automates what Passenger Assist does manually

Frees staff for cases needing human judgment

Ethics

- Data owned by the passenger are never shared, can be deleted at any time.
- Every guardian action is transparent and overridable
- Its fully welsh-oriented which is a rights-based commitment, not a feature

Risks

Connectivity issues

LCD retains journey info offline, buzzer alerts the passenger

Staff ignore alerts

Auto-escalates to station manager with every notification logged.

Weight and fit

It’s a lightweight universal clip, tested across wheelchair types

Cost Equity

It’s a subsidised model which ensures access is available for all

Our Emerging Technologies

Arduino

Physical Computing

ElevenLabs

Voice Interface

Cost and Funding Model

Hardware Per Unit

Production unit: est. £150 to £300

Infrastructure Cost

NFC upgrades extend existing hardware

Low marginal cost per station

Who Pays

TfW or NHS subsidy model

Issued like a disabled person's rail card where user doesn't pay.

The alternative: continuing to fund Passenger Assist staffing manually at higher long-term cost.

Heliwr in 2036 The 10-Year Vision

Fully Miniaturised

All processing in the module making it fit anywhere

FEIBR and cellular connectivity built in

National Standard

Extended beyond Wales across the UK

One guardian, one experience everywhere

Truly Personal AI

Learns passenger preferences over time

Becomes a genuine travel companion, not just a tool

Guardian

One Guardian, One Voice, One Journey.